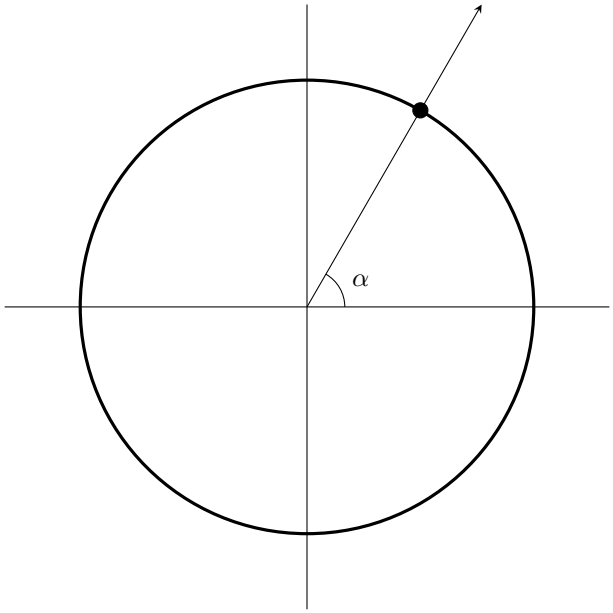
MATH 195: 10/20
CIRCLES, ANGLES, AND DIRECTIONS IN THE PLANE

Directions in the plane.

We can use angles to identify a point on a circle, which is the same as identifying a direction in two dimensional space. To write this as a number we need to pick three things.

- **A starting place.** We pick the positive x-axis to be angle 0.
- **A direction.** We pick counterclockwise to be the positive direction, clockwise to be the negative direction.
- **A scale.** We pick 2π to be the angular measure of a full rotation around the circle.



Radians.

For most mathematical purposes, radians are the most convenient measure of angles. The reason for this, is they have a direct meaning in terms of distances.

$$\theta \text{ radians} = \text{the angular measure which traces out a distance of } \theta \text{ many radii.}$$

The full circumference is 2π radians, which is why 2π is the measure of a full rotation.

Radians are a *ratio*, namely the ratio of arc length around the outside over the radius, so they are a dimensionless unit. We say, for example, an angle is $\pi/2$ without needing to specify radians. Compare to how an angle is 90° but not 90.

$$\text{Full circle} = 2\pi = 360^\circ$$
$$\text{Half circle} = \pi = 180^\circ$$
$$\text{Right angle} = \pi/2 = 90^\circ$$

Coterminal angles.

Many angles point the same direction. We call these *coterminal* angles. Two angles are coterminal if they differ by a whole number multiple of full rotations, that is if their difference is an integer multiple of 2π.

Quadrants and reference angles.

A lot of information about an angle is based on which *quadrant* it is in.

- Quadrant I: 0 to $\pi/2$, upper right.
- Quadrant II: $\pi/2$ to π, upper left.
- Quadrant III: π to $3\pi/2$, lower left.
- Quadrant IV: $3\pi/2$ to 2π, lower right.

For any angle, its *reference angle* is the magnitude of the smallest angular measure to get to the x-axis. This will be useful when doing trig. Note that reference angles are always between 0 and $\pi/2$.